Investigating Vanishing Gradients and Early-Layer Trainability Across Network Depth in Deep MLPs on BloodMNIST

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Abstract

Very small gradients can make early layers of deep neural networks difficult to train. This study examines how network depth and hidden-layer activation functions relate to weight-gradient attenuation and learning progress in multilayer perceptrons (MLPs) on BloodMNIST. In Stage A, Sigmoid MLPs with 2, 5, 10, and 20 hidden layers were compared. In Stage B, Sigmoid, Tanh, and ReLU were compared at five hidden layers. All models used Xavier Uniform initialization, plain SGD with learning rate 0.1, and 20 training epochs. The study included 18 unique main training runs across three random seeds. Measurements included layer-wise root-mean-square (RMS) weight gradients, relative gradient ratios, normalized first-layer weight displacement, training loss, and validation metrics. In Stage A, the mean first-to-last hidden-layer gradient ratio fell from 0.4070 (D2) to 2.711×10^{-12} (D20), alongside smaller first-layer weight displacement and slower training-loss reduction. At D5, Tanh and ReLU showed much less relative attenuation than Sigmoid; Tanh had the highest mean validation accuracy (86.21%) and macro-F1 (0.8441). These findings describe depth- and activation-dependent differences under the tested settings, without establishing a single causal mechanism.

1 Introduction

Training a neural network requires gradients to reach its parameters during backpropagation. In a deep network, gradients pass through many weight transformations and activation derivatives. Their magnitudes can become very small near the input, making it difficult for those layers to change during training. This is commonly called the *vanishing-gradient problem* (Bengio et al., 1994; Glorot and Bengio, 2010).

This report studies that problem in simple, constant-width MLPs rather than aiming to maximize blood-cell classification accuracy. BloodMNIST provides a standard image dataset for comparing architectures under a shared training protocol. Two research questions guide the work:

- **RQ1 (Primary Research Question):** How does increasing network depth affect layer-wise gradient propagation and the trainability of early layers in deep MLPs on BloodMNIST?

- **RQ2 (Secondary Research Question):** After a depth-dependent baseline is established, how do Sigmoid, Tanh, and ReLU differ in gradient attenuation, convergence behavior, and classification performance under controlled conditions?

The study examines three hypotheses:

H1 – Depth Effect. As network depth increases, gradients reaching layers closer to the input are expected to become smaller relative to later layers under settings susceptible to vanishing gradients.

H2 – Trainability Effect. Conditions showing stronger early-layer gradient attenuation are expected to exhibit smaller normalized early-layer weight changes and slower convergence. This relationship is evaluated using gradient attenuation, normalized first-layer weight displacement, training-loss trajectories, and epochs-to-threshold. The training-loss threshold was determined from the preliminary pilot experiment and fixed before the main comparisons.

H3 – Activation Effect. Under a common controlled initialization, Sigmoid and Tanh may show stronger depth-dependent gradient attenuation than ReLU; the magnitude of this difference may depend on initialization.

Experimental Scope. Stage A evaluates H1 and H2 by comparing Sigmoid MLPs with 2, 5, 10, and 20 hidden layers. Stage B evaluates activation-dependent differences at a fixed depth of five hidden layers (D5) using Sigmoid, Tanh, and ReLU under the same Xavier Uniform initialization and SGD training protocol.

Gradient attenuation is examined using layer-wise RMS weight gradients and first-to-last hidden-layer gradient ratios. Optimization progress is assessed using training-loss trajectories and the first epoch reaching the predefined loss threshold of 1.50, rather than a formal convergence criterion.

Stage B does not test activation-by-depth interactions or alternative initialization methods. Therefore, only the activation-dependent component of H3 is directly evaluated.

The main contribution is a reproducible empirical analysis connecting gradient attenuation, early-layer parameter displacement, and optimization progress under controlled experimental conditions.

2 Background and Related Work

2.1 Backpropagation and Gradient Attenuation

For hidden layer ℓ in an MLP, the forward calculation is

$$z^{(\ell)} = W^{(\ell)} a^{(\ell-1)} + b^{(\ell)}, \qquad a^{(\ell)} = \phi\Big(z^{(\ell)}\Big), \tag{1}$$

where W and b are the weight matrix and bias vector, and ϕ is the activation function. For a hidden layer followed by another linear transformation, backpropagation gives

$$\delta^{(\ell)} = \big(W^{(\ell+1)}\big)^{\top} \delta^{(\ell+1)} \odot \phi'\Big(z^{(\ell)}\Big). \tag{2}$$

Here $\delta^{(\ell)}$ is the loss gradient with respect to the layer's pre-activation. Multiplying through successive layers can make these gradients smaller. Bengio et al. (1994) studied related difficulties across time

in recurrent networks, while Glorot and Bengio (2010) examined activation and initialization effects in deep feedforward networks.

This study measures *weight gradients*, not the full backpropagated activation-gradient signal. Weight gradients also depend on the inputs to each layer, so their magnitudes are an informative but incomplete view of gradient propagation.

2.2 Activation Functions, Initialization, and Trainability

Sigmoid has derivative $\sigma'(z) = \sigma(z)(1-\sigma(z))$, which is at most 0.25 and approaches zero in saturated regions. Tanh is zero-centered, with derivative $1-\tanh^2(z)$; it can also saturate. ReLU has derivative 1 for positive inputs and 0 for negative inputs. These differences can influence gradients as they pass backward through the network (Glorot and Bengio, 2010; Glorot et al., 2011).

Initialization matters as well. Xavier (Glorot) initialization was designed to help balance signal magnitudes under simplifying assumptions (Glorot and Bengio, 2010). He initialization accounts for rectifier nonlinearities and can be more appropriate for ReLU networks (He et al., 2015). This experiment uses *Xavier Uniform with gain 1 for every activation* to keep initialization rule fixed, not because it is optimal for all three functions.

Schoenholz et al. (2017) related signal propagation and trainable depth through a mean-field framework. Their theoretical quantities are different from the RMS weight-gradient measurements used here. The present report therefore makes empirical comparisons rather than testing mean-field theory directly.

3 Methodology

3.1 Dataset and MLP Architecture

BloodMNIST is an eight-class biomedical image dataset in MedMNIST v2 (Yang et al., 2023), based on peripheral blood-cell imagery (Acevedo et al., 2020). Images are RGB and 28×28 pixels. The official partitions contain 11,959 training, 1,712 validation, and 3,421 test images. The official splits were kept unchanged, and no data augmentation was used. Inputs were scaled to $[0, 1]$ and normalized channel-wise using statistics computed from the *training split only*. The validation set was used for development evaluation; the test set was not used for final performance reporting.

Each image was flattened to $3 \times 28 \times 28 = 2{,}352$ inputs. The model contained D fully connected hidden layers with 256 units per layer, followed by a linear output layer with eight logits:

$$2352 \longrightarrow \underbrace{256 \longrightarrow \cdots \longrightarrow 256}_{D \text{ hidden layers}} \longrightarrow 8. \tag{3}$$

The same activation was applied after every hidden layer; no activation was applied to the output logits. There were no convolutional layers, normalization layers, dropout layers, or skip connections. Hidden width was fixed, but increasing depth also increased the number of parameters.

3.2 Experimental Design and Training

Stage A compared four Sigmoid models: D2, D5, D10, and D20, where D is the number of hidden linear layers. The objective was to observe how gradient magnitudes, first-layer parameter movement, and learning progress varied with depth.

Stage B compared Sigmoid, Tanh, and ReLU at D5. D5 was selected after Stage A because it showed clear early-layer attenuation but still made measurable learning progress. Its three Sigmoid runs were reused from Stage A instead of retrained. Stage A had 12 depth–seed runs; Stage B had nine activation–seed entries, including the three reused baselines. Thus, there were **18 unique main training runs** in total.

All main runs used the following protocol: Xavier Uniform initialization (gain 1) for all Linear weight matrices, zero biases, cross-entropy loss, SGD with learning rate 0.1, zero momentum, zero weight decay, batch size 128, 20 complete epochs, and random seeds 0, 1, and 2. A short three-epoch pilot on D2 and D20 (seed 0) showed no obvious numerical instability at learning rate 0.1. The learning rate and training-loss threshold of 1.50 were then fixed before the main experiments. No broad hyperparameter search was performed.

Each run used a newly initialized model and optimizer, and its own seeded training DataLoader. Gradients were recorded after `backward()` and before `optimizer.step()`. Validation loss, accuracy, and macro-F1 were measured after each epoch on the full validation split. No early stopping was used.

3.3 Gradient and Trainability Measurements

RMS weight gradients. For a hidden-layer weight-gradient tensor $G^{(\ell)}$ containing N_ℓ elements, RMS magnitude was defined as

$$g_\ell = \mathrm{RMS}(G^{(\ell)}) = \sqrt{\frac{1}{N_\ell}\sum_{i=1}^{N_\ell}\left(G_i^{(\ell)}\right)^2}. \tag{4}$$

Only hidden-layer *weight* gradients were included; biases and the output layer were excluded. The implementation used a rescaled norm to reduce numerical problems with very small values. For every epoch, the RMS gradients were averaged over its first 10 training mini-batches. The first 100 global training steps were also recorded for early-step inspection.

Relative gradient attenuation. The main relative measurement compared the first and last hidden layers:

$$R_e = \frac{\bar{g}_{1,e}}{\bar{g}_{D,e}}, \tag{5}$$

where $\bar{g}_{\ell,e}$ is the average sampled RMS gradient for layer ℓ in epoch e. A small R_e means that the first-layer weight gradient is much smaller than the last-layer weight gradient. The ratios were calculated *within each run* before averaging across seeds. Epoch-1 ratios were used for the main between-condition comparisons. They are not percentages of an activation-gradient signal preserved through backpropagation.

First-layer weight displacement. To describe how far the first hidden-layer weights moved from initialization, the study used

$$C_1(e) = \frac{\|W_1(e) - W_1(0)\|_F}{\|W_1(0)\|_F}. \tag{6}$$

This quantity was measured after each epoch; the epoch-20 value is used in the summary tables. It measures *net distance from initial weights*, not the sum of all SGD update magnitudes.

Training-loss milestone. The first epoch whose sample-weighted mean training loss was at most 1.50 was recorded as *epochs to threshold*. A run was marked *Not reached* if it did not meet the criterion within 20 epochs. Because training loss was accumulated while parameters were being updated across mini-batches, it is not the loss of a single fixed epoch-end model. Reaching the threshold describes learning progress, not mathematical convergence.

3.4 Statistical Reporting

For every condition, results were summarized across three seeds using the arithmetic mean and sample standard deviation ($n - 1$ denominator). Threshold-reaching epochs were summarized only among seeds that reached the threshold, and the number of successful seeds was shown separately. The validation macro-F1 was the unweighted mean of class-specific F1 scores across the eight classes. No confidence intervals or significance tests were computed.

4 Experimental Results

4.1 Stage A: Network Depth

Figure 1 compares gradients by hidden layer at epoch 1. As the Sigmoid network became deeper, gradients near the input became much smaller. The mean first-to-last RMS gradient ratio fell from 0.4070 for D2 to 0.006369 for D5, 4.837×10^{-6} for D10, and 2.711×10^{-12} for D20 (Table 1). Absolute first-layer RMS gradients also declined from 6.259×10^{-4} at D2 to 3.664×10^{-15} at D20.

Figure 2 follows D20 across 20 epochs. Its first-layer RMS gradient remained near 10^{-15} (epoch 1: 3.664×10^{-15}; epoch 20: 2.918×10^{-15}), whereas the last hidden layer remained near 10^{-3}. Severe early-layer attenuation therefore persisted throughout the observed training period.

The depth-related differences also appeared in optimization. After 20 epochs, mean normalized first-layer displacement was 0.3279 for D2, 0.1090 for D5, 2.740×10^{-5} for D10, and a *recorded* 0.0000 for D20. The D20 value should not be interpreted as proof of mathematically zero updates; finite-precision rounding may have contributed.

Figure 3 shows that D2 reduced training loss rapidly, D5 improved later, and D10 and D20 stayed near a loss of 2.01. All D2 runs reached the threshold at epoch 2; D5 reached it in epochs 14–15. D10 and D20 did not reach it within 20 epochs. Final validation accuracy decreased from 0.8123

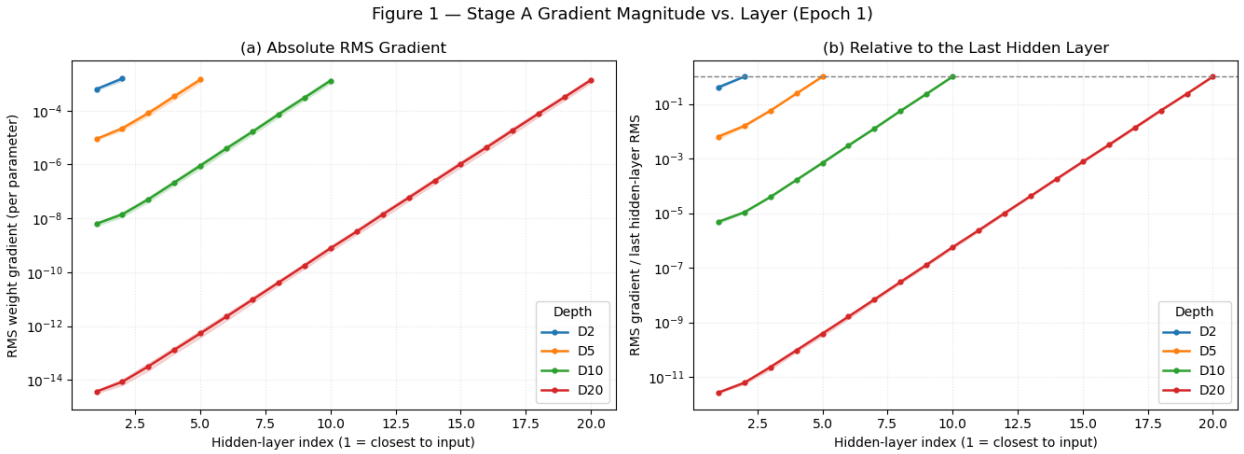


Figure 1: Layer-wise RMS weight gradients at epoch 1. (a) Absolute gradients and (b) gradients normalized by the last hidden layer, for D2, D5, D10, and D20 Sigmoid MLPs. Each epoch-level gradient uses the first 10 training batches. Curves are means across three seeds; shading denotes ±1 sample SD where visible on the logarithmic scale. Hidden Layer 1 is nearest the input.

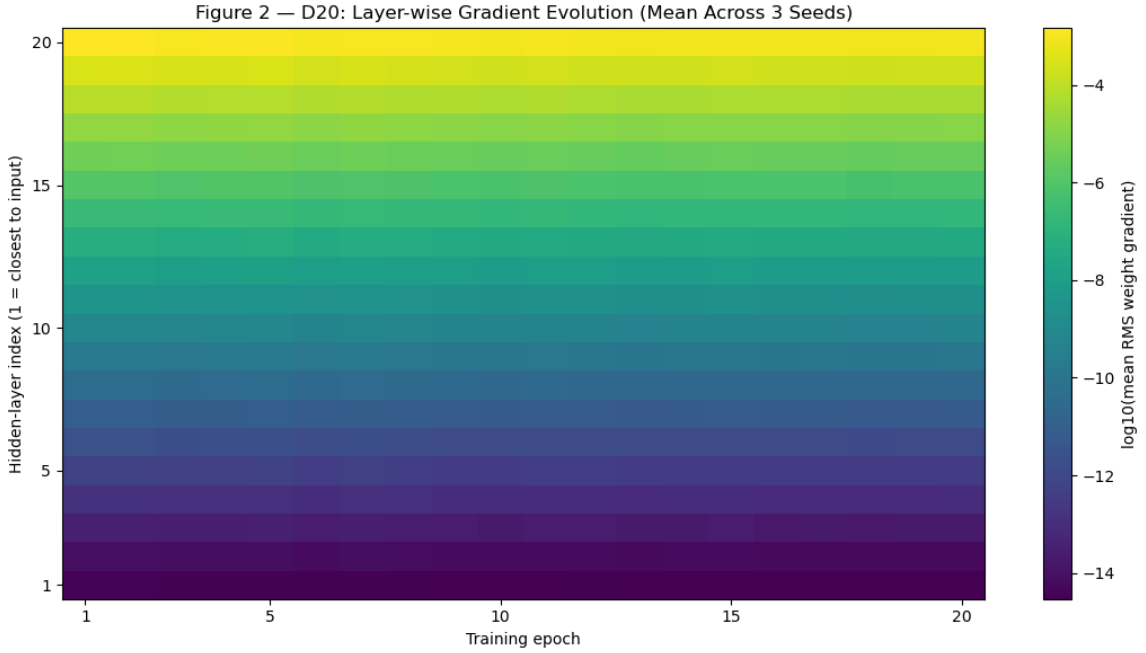


Figure 2: D20 gradient evolution. Heatmap of $\log_{10}$ of the mean RMS weight gradient across three seeds for each hidden layer and epoch. Each layer–epoch value uses the first 10 training batches. Hidden Layer 1 appears at the bottom. The colors display transformed means, not standard-deviation bands.

(D2) to 0.4663 (D5) and 0.1820 (D10 and D20). This performance pattern complements, but does not itself prove, gradient attenuation.

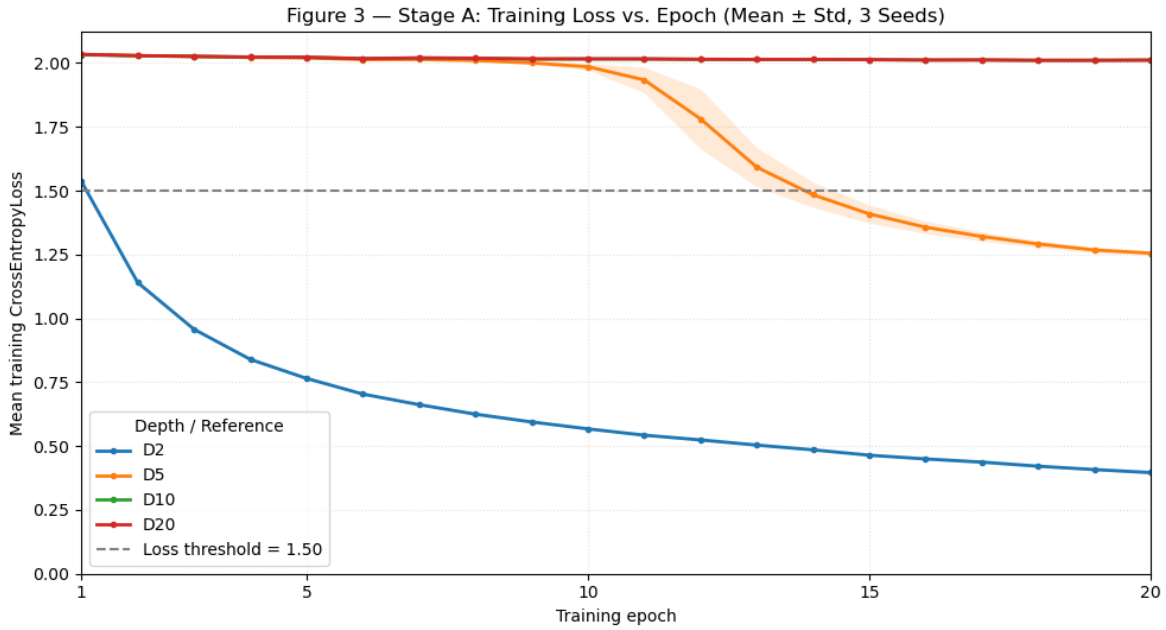


Figure 3: Training-loss trajectories. Mean training cross-entropy loss across three seeds over 20 epochs for D2, D5, D10, and D20 under Sigmoid–Xavier–SGD. Shaded bands represent ±1 sample SD. The dashed line marks the predefined loss threshold of 1.50. Epochs to threshold are calculated per run, not from the plotted mean curve.

Table 1: Stage A: depth comparison. Sigmoid activation, Xavier Uniform, SGD (learning rate 0.1), and 20 epochs. Values are mean ± sample SD across three seeds. E1 and E20 mean epochs 1 and 20.

Depth	Gradient ratio (E1)	W_1 change (E20)	Train loss (E20)
D2	$4.070\text{e}{-1} \pm 4.482\text{e}{-2}$	$3.279\text{e}{-1} \pm 2.546\text{e}{-3}$	0.3963 ± 0.0065
D5	$6.369\text{e}{-3} \pm 1.129\text{e}{-3}$	$1.090\text{e}{-1} \pm 4.262\text{e}{-3}$	1.2555 ± 0.0086
D10	$4.837\text{e}{-6} \pm 6.791\text{e}{-7}$	$2.740\text{e}{-5} \pm 8.056\text{e}{-6}$	2.0114 ± 0.0013
D20	$2.711\text{e}{-12} \pm 1.922\text{e}{-13}$	$0.000\text{e}{+0} \pm 0.000\text{e}{+0}$	2.0117 ± 0.0014

Depth	Epochs to threshold	Seeds reached	Val. accuracy	Val. macro-F1
D2	2.00 ± 0.00	3/3	0.8123 ± 0.0112	0.7851 ± 0.0208
D5	14.67 ± 0.58	3/3	0.4663 ± 0.0195	0.3377 ± 0.0213
D10	Not reached	0/3	0.1820 ± 0.0126	0.0385 ± 0.0022
D20	Not reached	0/3	0.1820 ± 0.0126	0.0385 ± 0.0022

Note. The predefined loss threshold is 1.50. "Not reached" means no seed met it during the 20-epoch budget. W_1 change is net normalized displacement from initialization.

4.2 Stage B: Activation Functions at D5

Stage B held depth at D5 and compared activation functions under the same training and initialization protocol. At epoch 1, the mean first-to-last gradient ratio was 0.006369 for Sigmoid, 0.4750 for Tanh, and 0.4954 for ReLU (Table 2). The corresponding absolute first-layer RMS gradients were 8.823×10^{-6}, 3.338×10^{-3}, and 1.991×10^{-3}. Tanh had the largest absolute first-layer gradient, while ReLU had the slightly larger *relative* ratio; these metrics describe different aspects of gradient magnitude.

Tanh and ReLU also produced more first-layer weight displacement and faster training-loss reduction than Sigmoid. All six Tanh/ReLU runs reached the threshold at epoch 1, compared with epochs 14–15 for Sigmoid. At epoch 20, Tanh had the highest mean validation accuracy (0.8621) and macro-F1 (0.8441), followed by ReLU (0.8170 and 0.7946) and Sigmoid (0.4663 and 0.3377). Relative gradient preservation and validation performance therefore did not produce identical rankings.

Table 2: Stage B: activation comparison at D5. Xavier Uniform, SGD (learning rate 0.1), and 20 epochs. Values are mean ± sample SD across three seeds. Sigmoid runs are reused from Stage A.

Activation	Gradient ratio (E1)	W_1 change (E20)	Train loss (E20)
Sigmoid	6.369e−3 ± 1.129e−3	1.090e−1 ± 4.262e−3	1.2555 ± 0.0086
Tanh	4.750e−1 ± 3.762e−2	3.981e−1 ± 5.189e−3	0.1934 ± 0.1173
ReLU	4.954e−1 ± 2.151e−2	4.033e−1 ± 3.480e−3	0.1992 ± 0.0765

Activation	Epochs to threshold	Seeds reached	Val. accuracy	Val. macro-F1
Sigmoid	14.67 ± 0.58	3/3	0.4663 ± 0.0195	0.3377 ± 0.0213
Tanh	1.00 ± 0.00	3/3	0.8621 ± 0.0079	0.8441 ± 0.0102
ReLU	1.00 ± 0.00	3/3	0.8170 ± 0.0322	0.7946 ± 0.0317

Note. Gradient ratios are calculated within runs before taking the mean across seeds. The training-loss threshold is 1.50.

5 Discussion

5.1 H1: Effect of Depth

Stage A strongly supports H1 *within the tested Sigmoid models*: increasing depth was accompanied by progressively smaller early-layer weight gradients relative to later layers. Figure 1 shows the pattern across depths, and Figure 2 shows that the extreme attenuation in D20 continued throughout training. Repeated multiplication by weights and activation derivatives is a plausible explanation, consistent with Glorot and Bengio (2010) and Schoenholz et al. (2017). However, this experiment did not measure activation saturation or Jacobian properties directly. It also did not hold total parameter count constant across depths.

5.2 H2: Attenuation and Early-Layer Trainability

H2 is supported as an *empirical association*. Conditions with stronger early-layer gradient attenuation also showed less first-layer weight displacement and slower training-loss reduction. This pattern appeared across depths in Stage A and across activations at fixed D5 in Stage B. Under plain SGD, smaller weight gradients imply smaller individual updates for a given learning rate, so the association is reasonable. But net displacement is not the same as the sum of updates, and other differences between models may affect training. The evidence does not show that gradient attenuation is the sole cause of poorer optimization.

5.3 H3: Effect of Activation Function

At D5, the data show a large difference between Sigmoid and the other two activations. Tanh and ReLU had substantially larger early-layer gradients, greater first-layer weight displacement, and faster training-loss reduction than Sigmoid. ReLU had a slightly higher mean relative gradient ratio than Tanh, but Tanh had better final validation performance. This distinction reinforces that gradient ratios and classification metrics answer different questions.

H3 is partially supported as originally formulated. The activation-dependent differences at D5 are clear in these runs, but Stage B did not compare activations across multiple depths or initialization schemes. The three-seed comparison also does not establish a reliable general ranking between Tanh and ReLU. Xavier Uniform was held constant for experimental control, not optimized separately for each activation (He et al., 2015).

6 Limitations and Future Work

The findings are limited to BloodMNIST, constant-width MLPs, and the chosen initialization and optimizer. Stage A changed parameter count as it changed depth; Stage B evaluated activations at D5 only. Every condition used three seeds and 20 epochs, so the results do not describe longer training or all possible initialization and optimization settings. Validation metrics were not followed by an independent test-set evaluation.

RMS weight gradients were averaged from the first 10 batches of each epoch rather than measured over every batch. These are parameter-gradient summaries, not direct measurements of activation-gradient signals or Jacobian singular values. The recorded D20 first-layer displacement of zero may reflect finite numerical precision. The predefined loss threshold measures attainment of a training milestone, not complete convergence. Finally, observed relationships among gradient attenuation, parameter displacement, and learning progress are associations rather than isolated causal effects.

Future work could compare activations across several depths under both Xavier and He initialization; examine activation saturation and inactive ReLU units; and test batch normalization or residual connections. Additional learning rates, optimizers, longer training, more seeds, and other biomedical image datasets would help assess how general the observed patterns are. These extensions are outside the current study.

7 Conclusion

This study used two controlled experimental stages to examine vanishing gradients and early-layer trainability in BloodMNIST MLPs. In Stage A, increasing Sigmoid-network depth from D2 to D20 was associated with much stronger early-layer RMS weight-gradient attenuation, smaller recorded first-layer displacement, and slower training-loss reduction. These results support H1 within the tested configuration and H2 as an empirical association.

In Stage B, Tanh and ReLU showed substantially less relative early-layer attenuation and faster optimization than Sigmoid at D5. Tanh had the highest observed mean validation accuracy and macro-F1, whereas ReLU had a slightly higher mean relative gradient ratio. The results partially support H3, but they do not establish activation-by-depth effects or sensitivity to alternative initialization. Taken together, the experiments show why gradient measurements, parameter displacement, and training outcomes should be examined together, while keeping conclusions within the evidence collected.

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